What Survives Diffusion Noise? Exact Scores, Order-by-Order Information Decay, and Transition Learning in Non-Gaussian Markov Chains

Anonymous Authors[1]

Abstract

Diffusion models are trained to approximate the score of a noised data distribution, a target that is unknown for almost every distribution of interest. We study a family in which it is known exactly and non-trivial: a first-order Markov chain with continuous, non-Gaussian transitions under coordinatewise Ornstein–Uhlenbeck noise, where conditioning leaves the posterior factor graph a tree and one forward–backward sweep of functional message passing returns the exact score. This laboratory lets us separate what diffusion noise does to different orders of structure. We prove that the channel attenuates every Hermite mode of degree m by e^{-mt}, so that the Fisher information one noised sequence carries about an order-m feature decays as e^{-2mt}: correlation (degree two) survives as e^{-4t}, non-Gaussian shape (degree four) as e^{-8t}, and the gap between the true score and that of the covariance-matched Gaussian—which we show is exactly what Gaussian message closure computes—is $\frac{1}{6}\|\kappa_4\|_F^2 e^{-8t}$, with the crossover set by the covariance spectrum. Tree-exact inference verifies the rates and the constants. We then learn the transition law from noised sequences alone, by expectation–maximisation around the same recursion with Fisher's identity supplying the exact gradient, and show that it recovers the fourth-order structure that unstructured and locality-aware denoisers trained on the clean data do not reach at the same number of sequences: the learned networks sit at or above the second-order floor 0.168 of the pooled score error, the structured estimator far below it. [PENDING: final capacity, baseline, generation and misspecification numbers]

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

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1. Introduction

Generative diffusion produces samples by reversing a noising process, and the reversal is driven by the score $S(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log P_t(\mathbf{x})$ of the noised data law (Sohl-Dickstein et al., 2015; Ho et al., 2020; Song et al., 2021). Training regresses a network onto that score through denoising (Vincent, 2011), and every question about what such a network has learned is a question about the distance between what it computes and the score it should have computed. For realistic data the target is unavailable, so a large error cannot be separated from a hard target; the tractable alternatives are Gaussian, whose scores are linear, or mixtures, whose nonlinearity lives in the mixture index rather than in any dependence between coordinates.

We study a family that is neither. A first-order Markov chain with continuous, non-Gaussian transitions has a globally dependent, nonlinear score, yet under coordinatewise Ornstein–Uhlenbeck noise its posterior factor graph is a tree, so the score is computable exactly by two sweeps of functional message passing (Section 2). Because the chain's covariance is fixed by two numbers and its higher-order structure by the innovation law, the model separates second-order from higher-order structure by construction: Gaussian, Laplace, Student-t, uniform and bimodal innovations, and a one-parameter family that interpolates between them, share one covariance exactly.

The question. What information about structured data survives diffusion noise, and how much does knowing the structure reduce the data needed to recover the score and the dynamics that generate it?

Diffusion noise is a low-pass filter on statistical order. The first answer is a theorem (Section 3). Under the

Ornstein–Uhlenbeck channel every Hermite mode of degree m is attenuated exactly by e^{-mt}, so the Fisher information a noised sequence carries about an order-m feature of the data decays as e^{-2mt}. Correlation lives at degree two and decays as e^{-4t}; non-Gaussian shape first appears at degree four and decays as e^{-8t}; and the gap between the true score and the score of the covariance-matched Gaussian—which we prove is exactly what single-Gaussian message closure computes (Proposition 1)—is $\frac{1}{6}\|\kappa_4\|_F^2 e^{-8t}$ to leading order, with an explicit crossover $t_* = \frac{1}{2}\log\lambda_{\max}(\mathbf{\Sigma}_0)$ before which the rates are shallower. The exact score lets us verify the exponents and the constants (Figure 2), and the result explains, rather than contradicts, earlier measurements on this model that stopped before t_*.

Learning through the channel. The second answer is empirical. We learn the transition law from noised sequences alone, by expectation–maximisation around the same exact recursion, with Fisher's identity supplying the exact gradient of the marginal likelihood from one forward–backward pass (Section 4). The theorem predicts that the shape of the innovation law is recovered e^{4t} times more slowly than its correlation at observation level t; the experiment confirms it, and locates the levels at which shape is still learnable. Against denoising networks trained on the clean trajectories—a fully connected network, a weight-shared window head, and dilated, recurrent, attention-based and multiscale architectures under one training recipe—the structured estimator recovers fourth-order structure that the networks do not reach at the same number of sequences: at every size tested the learned networks sit at or above the second-order floor of Proposition 1, and the structured estimator sits far below it (Section 5). We show that this advantage is earned by the non-Gaussian capacity of the innovation model and not only by the Markov factorisation, that it transfers to the generated distribution, that it extrapolates in sequence length, and that it has a measurable boundary under misspecification.

Contributions. (i) An exactly solvable non-Gaussian, correlated diffusion laboratory with a validated score reference, and the identification of Gaussian message closure with the LMMSE of the covariance twin (Proposition 1). (ii) Theorems 2 and 3: exact Hermite attenuation and the order-by-order decay of score gaps and Fisher information, with constants and crossover, verified with the exact score. (iii) Transition learning from noised data by exact-inference EM, with the predicted e^{4t} separation of shape from correlation recovery measured. (iv) A controlled comparison against learned denoisers—capacity attribution, fair architecture screen, information-matched training, generation, length extrapolation and misspecification—showing what explicit structure buys and where it stops.

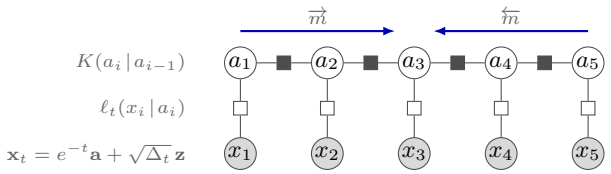


Figure 1. Why the score is exact. Coordinatewise noising attaches one unary factor per site to the chain prior and adds no edge among the latent variables, so the posterior factor graph is the prior's chain; two message sweeps return the posterior mean, and Tweedie's identity turns it into the score.

2. Setup: an exactly solvable non-Gaussian laboratory

Data and channel. A clean sequence $\mathbf{a} \in \mathbb{R}^L$ is a first-order Markov chain,

$$P_0(\mathbf{a}) = \mu(a_1)\prod_{i=2}^{L} K(a_i \mid a_{i-1}), \qquad K(a' \mid a) = \varphi(a' - \rho a), \tag{1}$$

with $a_1 \sim \mathcal{N}(0,1)$, $|\rho| < 1$, and an innovation density φ of mean zero and variance $q = 1-\rho^2$. The variance choice makes $\mathrm{Var}(a_i) = 1$ and $\mathrm{Cov}(a_i, a_j) = \rho^{|i-j|}$ exactly for every innovation law, so the families we use—Gaussian, Laplace, Student-t, uniform, a symmetric two-component mixture, and the generalised Gaussian $\varphi_\beta(\varepsilon) \propto \exp(-|\varepsilon/s_\beta|^\beta)$ whose exponent β moves the excess kurtosis from $+3$ (Laplace) through 0 (Gaussian) to -1.1 at fixed variance—share one covariance $\mathbf{\Sigma}_0$ and differ only through cumulants of order four and higher. The forward process is the coordinatewise Ornstein–Uhlenbeck channel

$$\mathbf{x}_t = e^{-t}\mathbf{a} + \sqrt{\Delta_t}\,\mathbf{z}, \qquad \Delta_t = 1 - e^{-2t}, \quad \mathbf{z} \sim \mathcal{N}(0, \mathbf{I}), \tag{2}$$

so $\mathrm{Var}(x_{t,i}) = 1$ at every t and the noised law P_t tends to $\mathcal{N}(0, \mathbf{I})$.

The score is a posterior mean, and the posterior is a tree. Tweedie's identity gives $S(\mathbf{x},t) = -(\mathbf{x} - e^{-t}\langle\mathbf{a}\rangle_{x,t})/\Delta_t$, with $\langle\mathbf{a}\rangle_{x,t} = \mathbb{E}[\mathbf{a} \mid \mathbf{x}_t = \mathbf{x}]$ (Robbins, 1956; Efron, 2011). Conditioning Equation (1) on $\mathbf{x}$ multiplies the prior by one unary factor $\ell_t(x_i \mid a_i) \propto \exp[-(x_i - e^{-t}a_i)^2/2\Delta_t]$ per site: no edge is added

among the latent variables, the posterior factor graph is the prior's chain, and sum-product on it is exact (Pearl, 1988; Kschischang et al., 2001). Two message sweeps,

$$\begin{aligned}\overrightarrow{m}_i(a) &\propto \int K(a \mid a')\,\ell_t(x_{i-1} \mid a')\,\overrightarrow{m}_{i-1}(a')\,\mathrm{d}a',\\ \overleftarrow{m}_i(a) &\propto \int K(a'' \mid a)\,\ell_t(x_{i+1} \mid a'')\,\overleftarrow{m}_{i+1}(a'')\,\mathrm{d}a'',\end{aligned} \tag{3}$$

return the site marginals $\propto \overrightarrow{m}_i \ell_t \overleftarrow{m}_i$ and hence the exact score of a correlated, non-Gaussian, L-dimensional law. Messages are functions on $\mathbb{R}$; we represent them on a grid of N_{g} points on $[-A, A]$ with trapezoidal quadrature, which makes the recursion exact for the discretised factor model and controlled for the continuous one (Appendix B: at $N_{\mathrm{g}} = 401$, $A = 8$ the reference is accurate to 10^{-5} relative on ordinary draws for every family used). Throughout, "the score" means this tree-exact functional inference, evaluated by validated grid quadrature.

The covariance twin and the price of second-order closure. The Gaussian chain with the same ρ and q has the closed-form score $S^{\mathrm{G}}(\mathbf{x}, t) = -\boldsymbol{\Sigma}_t^{-1}\mathbf{x}$, $\boldsymbol{\Sigma}_t = e^{-2t}\boldsymbol{\Sigma}_0 + \Delta_t \mathbf{I}$, and posterior mean $e^{-t}\boldsymbol{\Sigma}_0\boldsymbol{\Sigma}_t^{-1}\mathbf{x}$, the linear minimum-mean-square (LMMSE) estimator.

Proposition 1 (Gaussian closure computes the LMMSE of the covariance twin). Let Π_{G} project a density onto the Gaussian with its mean and variance, and let moment-projected sum-product apply Π_{G} to each message after every transition integral in Equation (3). For the chain Equation (1) with $\rho \neq 0$ and any innovation law of mean zero and variance q, it returns exactly the LMMSE estimator $e^{-t}\boldsymbol{\Sigma}_0\boldsymbol{\Sigma}_t^{-1}\mathbf{x}$, the same for every such law.

The proof (Appendix A) is an induction over the two sweeps: a convolution's first two moments depend on the innovation only through its first two moments, so every projected message equals the Gaussian chain's message. Proposition 1 turns the gap between S and S^{G} into an interpretable object: it is precisely what any estimator that carries only second-order structure—a Gaussian innovation kernel, a linear denoiser, or the Kalman smoother—must leave on the table. Section 3 computes its size.

3. What survives diffusion noise: attenuation by statistical order

Write γ for the standard Gaussian on $\mathbb{R}^L$, He_k for the probabilists' Hermite polynomials and $H_\nu(\mathbf{x}) = \prod_i \mathrm{He}_{\nu_i}(x_i)$ for their products, of degree $|\nu|$. The channel Equation (2) acts on Hermite moments diagonally.

Theorem 2 (Hermite attenuation). For every multi-index ν with $\mathbb{E}_{P_0}|\mathbf{a}|^{|\nu|} < \infty$, $\mathbb{E}_{P_t}[H_\nu(\mathbf{x}_t)] = e^{-|\nu|t}\,\mathbb{E}_{P_0}[H_\nu(\mathbf{a})]$.

Proof. Mehler's identity $\mathbb{E}_z[\mathrm{He}_n(\alpha a + \sqrt{1-\alpha^2}\,z)] = \alpha^n \mathrm{He}_n(a)$ for $z \sim \mathcal{N}(0,1)$ follows from the generating function $e^{sx - s^2/2} = \sum_n \mathrm{He}_n(x) s^n/n!$; coordinates are independent given $\mathbf{a}$, and $\alpha = e^{-t}$. □

Nothing here requires $P_0/\gamma \in L^2(\gamma)$—which fails for the Laplace chain and, because $\lambda_{\max}(\boldsymbol{\Sigma}_0) > 2$, for the Gaussian chain at $\rho = 0.85$—only finite moments. Two consequences are exact at every t: cumulants of order $m \geq 3$ scale as e^{-mt} (Gaussian noise has none), and the degree-4 Hermite moments of two variance-matched laws differ exactly by their fourth-cumulant tensors κ_4, attenuated by e^{-4t}. Correlation lives at degree two and decays as e^{-2t}; non-Gaussian shape first appears at degree four and decays as e^{-4t}. Diffusion noise is a low-pass filter on statistical order.

The consequence for scores and for learning is an expansion in the signal-to-noise ratio e^{-2t}. Let $s = e^{-t}\mathbf{x}/\Delta_t$, $b = e^{-2t}/\Delta_t$ and $\Lambda(s, b) = \log \mathbb{E}_{P_0} \exp(s \cdot \mathbf{a} - b|\mathbf{a}|^2/2)$, so that $\log P_t(\mathbf{x}) = \log \gamma_{\Delta_t}(\mathbf{x}) + \Lambda(s, b)$ and $S = -\mathbf{x}/\Delta_t + (e^{-t}/\Delta_t)\nabla_s \Lambda$ exactly. Grading Λ by order in e^{-t}, its second-order part is the same for P_0 and its covariance twin, and its fourth-order part differs by a term linear in κ_4.

Theorem 3 (Order-by-order information decay). Let P_0 have finite eighth moments and covariance $\boldsymbol{\Sigma}_0$, let S^{G} be the score of its covariance twin, and let $\lambda_{\max} = \lambda_{\max}(\boldsymbol{\Sigma}_0)$. As $t \to \infty$,

$$\mathbb{E}_{P_t}\|S - S^{\mathrm{G}}\|^2 = \tfrac{1}{6}\|\kappa_4\|_F^2\, e^{-8t}(1 + \epsilon_t), \tag{4}$$

$$I_t(\theta) = \tfrac{1}{2}\|\partial_\theta \boldsymbol{\Sigma}_0\|_F^2\, e^{-4t}(1 + \epsilon_t) \quad \text{if } \partial_\theta \boldsymbol{\Sigma}_0 \neq 0, \tag{5}$$

$$I_t(\theta) = \tfrac{1}{24}\|\partial_\theta \kappa_4\|_F^2\, e^{-8t}(1 + \epsilon_t) \quad \text{if } \partial_\theta \boldsymbol{\Sigma}_0 = 0, \tag{6}$$

the last with $\partial_\theta \kappa_4 \neq 0$, and with $\epsilon_t = O(e^{-2t}\lambda_{\max})$, where $I_t(\theta) = \mathbb{E}_{P_t}[(\partial_\theta \log P_t)^2]$ is the Fisher information one noised sequence carries about a parameter θ of P_0, and $\|\cdot\|_F$ is the Frobenius norm of the matrix or symmetric 4-tensor. For a Gaussian chain Equation (5) is exact at every t with $I_t = \frac{1}{2}\operatorname{tr}(\boldsymbol{\Sigma}_t^{-1}\partial_\theta \boldsymbol{\Sigma}_t \boldsymbol{\Sigma}_t^{-1}\partial_\theta \boldsymbol{\Sigma}_t)$.

The information statements Equations (5) and (6) are the diffusion-time form of a principle established for low-SNR latent-variable models: the log-likelihood expands in powers of the SNR with the order-k term carrying k-th-moment information (Katsevich & Bandeira, 2023; Perry et al., 2019; Bandeira et al., 2023),

and Fisher information about directions first identifiable at moment order k scales as SNR^k (Balanov et al., 2026); with $\mathrm{SNR}_t \simeq e^{-2t}$ these are Equations (5) and (6). The cumulant attenuation behind Theorem 2 is also written down for the forward process by Aarts et al. (2025). What is new is the score-level statement Equation (4) with its constant, made interpretable by Proposition 1; the crossover below; and the exact verification and learning consequences that follow. The proof (Appendix A) expands Λ to fourth order, takes the γ-second moment of the leading Hermite term by Wick's theorem, and bounds the remainder by the moment condition; the scalar leading term of Equation (4) is $(\kappa_4/6)\mathrm{He}_3(x)e^{-4t}$. The leading orders of a degree-2 and a degree-4 coordinate do not couple, because He_2 and He_4 are γ-orthogonal, so the efficient information of a shape coordinate after the nuisance coordinates are estimated has the same rate as its raw information.

Remark 4 (General order). The same grading gives, for a coordinate whose lowest non-vanishing Hermite footprint has degree m, $I_t(\theta) \propto e^{-2mt}$: the leading term of $\partial_\theta \log P_t$ is e^{-mt} times a degree-m Hermite polynomial, whose γ-second moment is a constant. The constants are explicit for $m = 2, 4$ above.

Corollary 5 (Sample complexity by order). Estimating a covariance coordinate to fixed precision from sequences observed at level t needs $M \propto e^{4t}$ of them; a shape coordinate needs $M \propto e^{8t}$. The ratio of the two informations is $\propto e^{-4t}$.

Where the asymptote begins. The expansion parameter is $e^{-2t}\lambda_{\max}(\mathbf{\Sigma}_0)$, not e^{-2t}, with $\lambda_{\max} \le (1+\rho)/(1-\rho)$ for the chain. Below $t_* = \frac{1}{2}\log\lambda_{\max}(\mathbf{\Sigma}_0)$ the long-wavelength covariance modes are still of order one and the rates are shallower; above it the exponents of Theorem 3 hold. At $\rho = 0.85$, $t_* = 1.17$: a strongly correlated chain keeps its correlation information longer, and any measurement stopped near t_* sees slopes short of -4 and -8.

Verification with the exact score. Figure 2 tests Theorem 3 where nothing has to be approximated. For the Gaussian chain the closed form gives slope -3.97 above t_* at $\rho = 0.85$ and the scaled information 0.24, 0.69, 0.95 at $t = 1, 2, 3$. For the non-Gaussian chain the shape information, estimated from per-sequence scores obtained by Fisher's identity through the tree-exact recursion, falls with slope -7.94 ($\beta = 2$) and -8.01 ($\beta = 3$) above t_*, against the predicted -8, while the correlation information falls with slope -3.98 against -4; the kinked Laplace density ($\beta = 1$) reaches the quadrature floor of the outer-product estimate near $t = 2.5$. The closure gap on the Laplace chain decays with slope -7.93 and reproduces the constant $\frac{1}{6}\|\kappa_4\|_F^2$ (113.0 for $L = 32$) to 0.99 at $t = 3$. The theorem also explains a measurement the previous literature on this model reported as a discrepancy: information slopes of about -3 and -6 obtained over $t \le 1.6$ are the pre-asymptotic values below $t_* = 1.17$, not a failure of the e^{-4t}, e^{-8t} law.

What this says about score learning. An estimator that carries only second-order structure cannot go below the LMMSE floor $\|S - S^{\mathrm{G}}\|$, which on the Laplace chain at $\rho = 0.85$ is 0.168 of the score energy pooled over the twelve-level schedule and 0.28 at the lowest level (Appendix D). A learner that recovers the fourth-order structure has, by Corollary 5, e^{4t} fewer effective samples per sequence at level t than one that recovers only covariance: the twelve-level schedule is, for shape, effectively a four-level one. Both facts organise the experiments of Section 5.

4. Learning the dynamics through noise

Now drop the assumption that K is known. We observe M noised sequences $\mathbf{x}^\mu$, each at one level t_μ, never the clean chain, and maximise the marginal log-likelihood $\mathcal{L}(\theta) = \sum_\mu \log P_t[t_\mu, \theta](\mathbf{x}^\mu)$ of a parameterised kernel K_θ.

The exact gradient from one sweep. Because only the transition factors depend on θ, Fisher's identity (Cappé et al., 2005) gives, on the grid,

$$\nabla_\theta \mathcal{L}(\theta) = \big\langle \mathbf{\Xi}(\theta),\, \nabla_\theta \log K_\theta \big\rangle, \qquad \mathbf{\Xi}_{kl} = \sum_\mu \sum_{i \ge 2} P(a_i = u_k, a_{i-1} = u_l \mid \mathbf{x}^\mu), \tag{7}$$

where the expected transition mass $\mathbf{\Xi}$ is read off the pairwise marginals that the same forward–backward pass returns. No differentiation through the recursion is needed, and the identity is verified against finite differences to 10^{-9} (Appendix C). $\mathbf{\Xi}$ is also all the M-step of EM needs: for a Gaussian kernel the maximiser is the belief-weighted Yule–Walker pair $\rho = S_{01}/S_{00}$, $q = (S_{11} - \rho S_{01})/W$ with $S_{ab} = \sum_{kl} \mathbf{\Xi}_{kl} u_k^a u_l^b$; for the mixture-innovation kernel $K_\theta(a' \mid a) = \sum_{c=1}^{C} \pi_c \mathcal{N}\big(a' - \rho a - \nu_c, s_c^2\big)$ used below, the complete-data problem carries a second latent label and we take conditional-maximisation sweeps, a generalised EM whose ascent is monotone (Dempster et al., 1977; Meng & Rubin, 1993). The estimator sees one frozen noised view per sequence and never a clean trajectory; each reported fit selects its iteration count on a validation bundle disjoint from the test bundle.

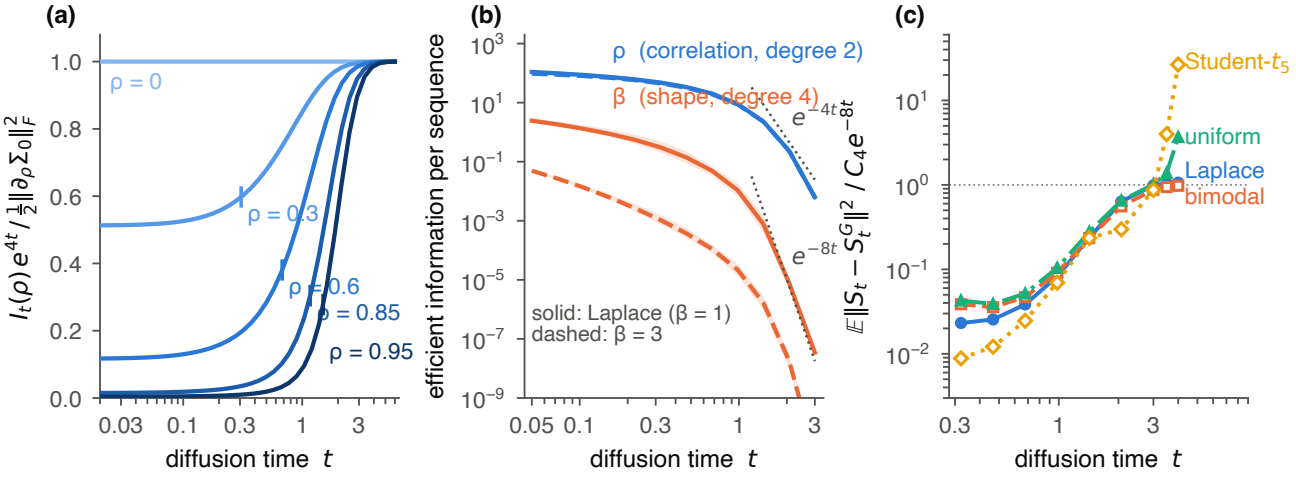


Figure 2. What survives diffusion noise. (a) Exact Fisher information of the Gaussian chain about ρ, scaled by its predicted asymptote $\frac{1}{2}\|\partial_\rho \mathbf{\Sigma}_0\|_F^2 e^{-4t}$, for five ρ; every curve reaches 1, and the tick on each marks $t_*(\rho)$. (b) Efficient information per noised sequence about the correlation coordinate ρ (degree 2) and the shape coordinate β (degree 4) of the generalised-Gaussian chain, from tree-exact inference and Fisher's identity, 512 sequences $\times$ 8 replicates (bands: replicate range); guides e^{-4t} and e^{-8t}. Above t_* the fitted slopes are -4.00 (ρ) and -8.01 (β, at $\beta = 3$). (c) The price of second-order closure: $\mathbb{E}\|S - S^{\mathrm{G}}\|^2$ measured by tree-exact BP, divided by the predicted $\frac{1}{6}\|\kappa_4\|_F^2 e^{-8t}$, for four innovation families. Laplace and the bimodal mixture reach the constant to within 0.99–0.98; Student-t_5, whose sixth moment is infinite, departs as the moment condition predicts; the uniform family reaches the floor of double precision near $t = 3.5$.

[PENDING: fig_learning.pdf (E06)]

Figure 3. Learning through noise. Recovery of the correlation coordinate and of the innovation shape from $M = 512$ sequences observed at a single level t, sixteen replicates; the predicted growth e^{2t} and e^{4t} of the two errors beyond t_* (dotted).

Shape is learned e^{4t} more slowly than correlation. Corollary 5 makes a prediction about learning that the exact score does not need: fitting from M sequences all observed at level t, the error of $\widehat{\rho}$ should grow like e^{2t} and the error of the fitted innovation excess kurtosis $\widehat{\kappa_4}$ like e^{4t} beyond t_*. Figure 3 measures both with $M = 512$ Laplace-chain sequences and 16 replicates per level. [PENDING: E06 queued: slopes of the two RMSEs against the level] This is the mechanism behind the convergence asymmetry reported for this model—correlation settles in tens of EM updates, shape in hundreds—and behind the one-view protocol's arithmetic: of twelve levels spread over $t \in [0.05, 3]$, only the lowest four carry appreciable information about shape.

5. What explicit structure buys, and where it stops

All comparisons use the frozen configuration of Appendix C: Laplace chain, $\rho = 0.85$, $L = 32$, twelve log-spaced levels on $[0.05, 3]$, and the schedule-pooled relative score error against the tree-exact reference (squared error and reference energy summed over levels before the square root). Sixteen paired seeds are the inferential unit; architectures and hyperparameters are selected on four development seeds that never enter a reported number, and every arm's budget is selected on a per-seed validation bundle disjoint from the test bundle.

The oracle ladder. Three reference points bracket every learned estimator (Appendix D): the isotropic score $-\mathbf{x}$, which knows only the marginal variance, at pooled error 0.670; the LMMSE oracle with the true covariance, at 0.168, which by Proposition 1 is the floor of every second-order estimator; and the exact score at zero.

Capacity: the margin is fourth-order. [PENDING: E02a running: capacity table and paired differences] A Gaussian-innovation kernel ($C = 1$) lands on the LMMSE floor, as Proposition 1 requires; the mixture kernel goes far below it. An earlier study of this model concluded the opposite from a sweep whose single-Gaussian arm alone stopped at its validation optimum while richer arms ran to a fixed cap; with the checkpoint selected identically for every capacity the effect is unambiguous (Appendix E). [PENDING: E02b: the generalised-Gaussian family, $\beta \in \{1, \ldots, 6\}$: the exact closure gap, the $C = 8$ vs 1 gap, and the window head's error against κ_4]

Baselines under one recipe. [PENDING: E03 pending: screen and confirmatory table; information-matched arm; length extrapolation]

Generation. [PENDING: E05 pending: reverse-SDE samples under the exact, EM–BP, LMMSE and network scores]

Misspecification. [PENDING: E08 pending: advantage against a misspecification parameter; break-even]

6. Related work

[PENDING: Rewritten from docs/RELATED_WORK_SWEEP.md once the sweep is verified.] Diffusion and score matching are due to Sohl-Dickstein et al. (2015); Ho et al. (2020); Song et al. (2021) and Hyvärinen (2005); Vincent (2011); the score–posterior-mean identity is Tweedie's (Robbins, 1956; Efron, 2011). Solvable settings and the statistical mechanics of diffusion (Biroli & Mézard, 2023; Biroli et al., 2024; Cui et al., 2025; Raya & Ambrogioni, 2023; Ghio et al., 2024) analyse memorisation and speciation with targets that are Gaussian or mixture-like; hierarchical models supply tractable non-Gaussian families (Sclocchi et al., 2025) on which U-Nets and transformers implement belief propagation (Mei, 2024; Garnier-Brun et al., 2024). Our chain is the degenerate tree of depth L, and our estimator is classical: sum-product on a chain is forward–backward (Baum et al., 1970; Rabiner, 1989), reducing to Kalman–RTS in the Gaussian case; grid quadrature is Kitagawa's smoother (Kitagawa, 1987); Fisher's identity and EM are Dempster et al. (1977); Cappé et al. (2005). Theorem 3 belongs to the low-SNR literature. In Gaussian channels the mutual information and the MMSE expand in the SNR with fourth-order statistics entering at second order (Prelov & Verdú, 2004; Guo et al., 2005; Dytso et al., 2023); for latent-variable models the log-likelihood expands with the order-k term matching k-th moments (Katsevich & Bandeira, 2023), which gives the σ^{2k} sample complexity of multireference alignment and orbit recovery (Perry et al., 2019; Bandeira et al., 2023) and, most closely, the layered Fisher-information geometry of Balanov et al. (2026). Aarts et al. (2025) write the forward-process cumulant attenuation e^{-nt} and show that score networks regenerate the cumulants. Wang & Vastola (2024) and Li et al. (2024) observe that learned scores are dominated by their Gaussian part over a wider noise range than a first-order bound suggests; Equation (4) is the sharp rate behind that observation for a non-Gaussian chain, where the Gaussian part is exactly the Kalman smoother. New here are the score-level statement with its constant and crossover, the Markov (dependent) rather than i.i.d. latent structure, the exact verification, and the learning consequences with an exact E-step rather than moment matching.

7. Limitations and conclusion

The model is a chain in one dimension with a linear autoregressive form; the theorem holds for any law with the stated moments, but the exact verification and the learning results are for chains. The estimator's advantage is the correctness of its factorisation: under a non-Markov truth it is misspecified in a way more data does not repair, and Section 5 locates where that begins. Inference costs $O(LN_g^2)$ per sequence, far above a network forward pass, so the advantage is in data and understanding, not in wall-clock. The comparison with denoisers equalises the number of clean sequences under each method's standard recipe and, separately, matches the information interface; it does not rank learning algorithms in general.

What the laboratory teaches is general: diffusion noise erases statistical structure order by order, at rates fixed by Hermite degree, so a score learner that reaches only second order is bounded away from the truth by an e^{-8t}-scaled term and a learner of shape pays e^{4t} in samples per level. Explicit structure is the way to recover what the data still carry; the price of the wrong structure is a bias that no sample size removes.

Impact statement

This paper presents theoretical and empirical work on how diffusion noise erases statistical structure and on learning structured dynamics from noised data, evaluated on synthetic models with known ground truth. Its societal consequences are those of a better understanding of generative diffusion models in general; we see no specific ethical concern that requires discussion here.

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A. Proofs

A.1. Proposition 1

Write $\overrightarrow{m}_i^{\mathrm{p}}$ for the forward message at site i after projection and ℓ_i for the Gaussian unary likelihood. We show by induction that every projected message equals the corresponding exact message of the covariance-matched Gaussian chain. The initial message is Gaussian or projected to the Gaussian with the same first two moments, which is the Gaussian chain's initial message. If $\overrightarrow{m}_{i-1}^{\mathrm{p}}$ is that chain's message, with mean m and variance v, multiplying by ℓ_{i-1} gives a Gaussian with the precision-weighted moments, identical in both chains; the transition integral against $a_i = \rho a_{i-1} + \varepsilon_i$ is a convolution whose mean and variance are $(\rho m, \rho^2 v + q)$ for every innovation with moments $(0, q)$, so Π_{G} returns the Gaussian chain's next message. The backward sweep runs the same argument with the inverted map $(m, v) \mapsto (m/\rho, (v+q)/\rho^2)$, which is where $\rho \neq 0$ is used, starting from the terminal likelihood ℓ_L (the flat terminal message is not normalisable and contributes no factor). Every projected belief is therefore the Gaussian chain's belief, whose posterior mean is linear in $\mathbf{x}$ and coincides with the LMMSE estimator $e^{-t}\mathbf{\Sigma}_0\mathbf{\Sigma}_t^{-1}\mathbf{x}$. Since the recursion never read φ beyond $(0, q)$, the estimator is the same for every such law. The algebraic steps (the two moment maps as integrals, the covariance identities, and the optimality of the normal-equation gain in arbitrary dimension) are machine-checked in Lean 4 (lean/README.md); the projection step itself is not.

A.2. Theorem 2

For $z \sim \mathcal{N}(0,1)$ and real a, α with $\beta = \sqrt{1-\alpha^2}$, the generating function $e^{sx - s^2/2} = \sum_n \mathrm{He}_n(x) s^n/n!$ evaluated at $x = \alpha a + \beta z$ and averaged over z gives $e^{s\alpha a - s^2\alpha^2/2} = \sum_n \mathrm{He}_n(a)(\alpha s)^n/n!$, hence $\mathbb{E}_z \mathrm{He}_n(\alpha a + \beta z) = \alpha^n \mathrm{He}_n(a)$. The coordinates of $\mathbf{z}$ are independent given $\mathbf{a}$, so the product form holds for H_ν; integrating over $\mathbf{a} \sim P_0$ requires only $\mathbb{E}|\mathbf{a}|^{|\nu|} < \infty$. The cases $n \leq 4$ are verified in Lean as polynomial identities under the first four Gaussian moments.

A.3. Theorem 3

Exact identities. With $\alpha = e^{-t}$ and $\Delta_t = 1 - \alpha^2$, Bayes' rule gives $P_t(\mathbf{x}) = \gamma_{\Delta_t}(\mathbf{x})\, \mathbb{E}_{P_0} \exp\!\big(\alpha \mathbf{x}\cdot\mathbf{a}/\Delta_t - \alpha^2|\mathbf{a}|^2/2\Delta_t\big)$, where γ_{Δ_t} is the $\mathcal{N}(0, \Delta_t \mathbf{I})$ density. Hence $\log P_t(\mathbf{x}) = \log \gamma_{\Delta_t}(\mathbf{x}) + \Lambda(s, b)$ with $s = \alpha\mathbf{x}/\Delta_t$, $b = \alpha^2/\Delta_t$ and $\Lambda(s,b) = \log \mathbb{E}_{P_0} \exp(s\cdot\mathbf{a} - b|\mathbf{a}|^2/2)$. For $b > 0$ the tilted measure is integrable for every s, so Λ is real-analytic in (s, b) on $\mathbb{R}^L \times (0, \infty)$, and its Taylor coefficients at the origin are the joint cumulants of the linear form $s\cdot\mathbf{a}$ and the quadratic form $|\mathbf{a}|^2/2$; $\mathbb{E}[\mathbf{a} \mid \mathbf{x}] = \nabla_s \Lambda$ and $S = -\mathbf{x}/\Delta_t + (\alpha/\Delta_t)\nabla_s \Lambda$.

Grading. Expand Λ in (s, b) at the origin and grade a monomial $s^j b^k$ by $j + 2k$; since $s = O(\alpha)$ and $b = O(\alpha^2)$, the grade is the order in α. By the symmetry of P_0 under $\mathbf{a} \mapsto -\mathbf{a}$ only even grades occur: $\Lambda = \Lambda_2 + \Lambda_4 + R_6$, with $\Lambda_2 = \frac{1}{2} s^\top \mathbf{\Sigma}_0 s - \frac{b}{2} \operatorname{tr} \mathbf{\Sigma}_0$ and

$$\Lambda_4 = \tfrac{1}{24}\kappa_4[s,s,s,s] - \tfrac{b}{4}\sum_i \operatorname{cum}(s\cdot\mathbf{a}, s\cdot\mathbf{a}, a_i, a_i) + \tfrac{b^2}{8}\sum_{ij} \operatorname{cum}(a_i, a_i, a_j, a_j),$$

where cum is the joint cumulant under P_0. Every fourth-order joint cumulant of linear forms in $\mathbf{a}$ equals its value under the covariance twin plus a linear contraction of κ_4, so $\Lambda_4 - \Lambda_4^{\mathrm{G}}$ is linear in κ_4 and $\Lambda_2 = \Lambda_2^{\mathrm{G}}$. The remainder R_6 is the Taylor remainder of grade six; on $\{|\alpha\mathbf{x}| \leq 1\}$ it is bounded by $C\alpha^6(1 + |\mathbf{x}|)^6$ with C depending on the moments of P_0 up to order eight.

Leading term of the score gap. $S - S^{\mathrm{G}} = (\alpha/\Delta_t)\nabla_s(\Lambda - \Lambda^{\mathrm{G}}) = (\alpha/\Delta_t)\big[\nabla_s(\Lambda_4 - \Lambda_4^{\mathrm{G}}) + \nabla_s R_6 - \nabla_s R_6^{\mathrm{G}}\big]$. Substituting $s = \alpha\mathbf{x}/\Delta_t$, $b = \alpha^2/\Delta_t$ and expanding in α, the grade-four term is $\alpha^4\, \nabla_{\mathbf{x}}\big[\frac{1}{24}\sum_{ijkl}\kappa_4[ijkl] :x_i x_j x_k x_l:\big] + O(\alpha^6(1+|\mathbf{x}|)^5)$, where $:\cdot:$ is the Wick (Hermite) product: the three contributions of Λ_4 combine into the Hermite form because $b = \alpha^2(1 + O(\alpha^2))$ and the cumulant contractions with δ_{ii} are exactly the terms that convert monomials into Hermite polynomials. The m-th component is $\alpha^4 \frac{1}{6}\sum_{jkl}\kappa_4[mjkl] :x_j x_k x_l:$.

Second moment. Under γ, Wick's theorem gives $\mathbb{E}_\gamma[:x_j x_k x_l: :x_{j'} x_{k'} x_{l'}:]$ equal to the number of perfect matchings of $\{j,k,l\}$ with $\{j',k',l'\}$ respecting indices; contracted with the symmetric tensor this yields

$\mathbb{E}_\gamma\big[(\frac{1}{6}\sum_{jkl}\kappa_4[mjkl]{:}x_jx_kx_l{:})^2\big] = \frac{3!}{36}\sum_{jkl}\kappa_4[mjkl]^2$, and summing over m, $\mathbb{E}_\gamma\|\text{leading term}\|^2 = \frac{1}{6}\|\kappa_4\|_F^2\alpha^8$. Replacing $\mathbb{E}_\gamma$ by $\mathbb{E}_{P_t}$ costs a factor $1+O(\alpha^2\lambda_{\max})$: on the bulk, $P_t/\gamma = 1+\alpha^2\frac{1}{2}\langle\mathbf{\Sigma}_0-\mathbf{I}, {:}\mathbf{x}\mathbf{x}^\top{:}\rangle + O(\alpha^4\lambda_{\max}^2(1+|\mathbf{x}|)^4)$, and the degree-2 correction has zero γ-mean against the square of a degree-3 polynomial up to terms of relative size $\alpha^2\lambda_{\max}$. The cross term between the leading term and the grade-six remainder is $O(\alpha^{10}\lambda_{\max})$ by Cauchy–Schwarz and the moment bound.

Tail. On $\{|\alpha\mathbf{x}| > 1\}$ the expansion is not used. There $|S - S^{\mathrm{G}}| \le 2|\mathbf{x}|/\Delta_t + \alpha(\mathbb{E}[|\mathbf{a}|\,|\mathbf{x}] + |\mathbb{E}^{\mathrm{G}}[\mathbf{a}|\mathbf{x}]|)/\Delta_t \le C(1+|\mathbf{x}|)/\Delta_t$ because the tilted mean is bounded by a polynomial in $|\mathbf{x}|$ under the moment condition, while $P_t(|\alpha\mathbf{x}| > 1) \le \exp(-c/\alpha^2)$ since $\mathbf{x}_t$ is a Gaussian convolution with variance at most one per coordinate. The tail contribution is therefore $O(\exp(-c'e^{2t}))$, below every power of α. Together, $\mathbb{E}_{P_t}\|S-S^{\mathrm{G}}\|^2 = \frac{1}{6}\|\kappa_4\|_F^2 e^{-8t}(1+O(e^{-2t}\lambda_{\max}))$, which is Equation (4).

Information. $\partial_\theta \log P_t = \partial_\theta\Lambda$. If $\partial_\theta\mathbf{\Sigma}_0 \neq 0$ the grade-two term $\frac{\alpha^2}{2}\langle\partial_\theta\mathbf{\Sigma}_0, {:}\mathbf{x}\mathbf{x}^\top{:}\rangle$ leads, and $\mathbb{E}_\gamma$ of its square is $\frac{\alpha^4}{4}\cdot 2\|\partial_\theta\mathbf{\Sigma}_0\|_F^2$ by Wick, giving Equation (5). If $\partial_\theta\mathbf{\Sigma}_0 = 0$ the grade-four term $\frac{\alpha^4}{24}\sum\partial_\theta\kappa_4[ijkl]{:}x_ix_jx_kx_l{:}$ leads, with γ-second moment $\frac{4!}{24^2}\|\partial_\theta\kappa_4\|_F^2\alpha^8$, giving Equation (6); the corrections and the tail are handled as above. Since ${:}\mathbf{x}\mathbf{x}^\top{:}$ and the degree-four Hermite polynomials are γ-orthogonal, the leading cross-information between a degree-2 and a degree-4 coordinate vanishes, so the leading term of the efficient information of the degree-4 coordinate equals that of its raw information. For the Gaussian chain $\mathbf{x}_t \sim \mathcal{N}(0, \mathbf{\Sigma}_t(\theta))$ exactly and the stated trace formula is the Gaussian Fisher information; its limit $e^{4t}I_t \to \frac{1}{2}\|\partial_\theta\mathbf{\Sigma}_0\|_F^2$ follows from $\mathbf{\Sigma}_t \to \mathbf{I}$ and $\partial_\theta\mathbf{\Sigma}_t = \alpha^2\partial_\theta\mathbf{\Sigma}_0$, and is machine-checked in Lean (lean/README.md, target 3).

The crossover. The expansion of P_t/γ used above is an expansion in $\alpha^2\mathbf{\Sigma}_0$, i.e. in $\alpha^2\lambda_{\max}(\mathbf{\Sigma}_0)$ in operator norm; for the chain $\lambda_{\max} \le (1+\rho)/(1-\rho)$ by the Toeplitz bound $\lambda_{\max} \le \sum_k \rho^{|k|}$. This is the source of the factor $\lambda_{\max}$ in every remainder and of $t_* = \frac{1}{2}\log\lambda_{\max}$; for the Gaussian chain the exact formula shows the same scale directly, since $\mathbf{\Sigma}_t^{-1}$ is within $\alpha^2\lambda_{\max}$ of $\mathbf{I}$ in operator norm.

B. Numerical representation and its control

Messages are represented on a uniform grid of N_{g} points on $[-A, A]$ with composite trapezoidal weights; the recursion is a scaled linear-domain forward–backward whose discarded constants are accumulated so the evidence $\log P_t(\mathbf{x})$ of the discretised model is exact to floating point. Three levels of exactness are kept apart: sum-product is exact at the level of functional messages; forward–backward is exact for the discretised factor model; and that model approximates the continuous chain with truncation (mass outside $[-A, A]$) and quadrature (second order in the spacing) errors that are measured, not assumed.

Validation where the answer is known. At $N_{\mathrm{g}} = 401$, $A = 8$ the recursion reproduces the Gaussian closed form and brute-force enumeration of the discretised posterior on short chains to double precision (tests/test_gaussian_bp_equivalence.py, tests/test_fisher_bruteforce.py). Against a finer reference ($N_{\mathrm{g}} = 1601$, $A = 12$) the worst relative score error over the twelve levels and four innovation families on ordinary draws is $9.9e-04$; under a three-standard-deviation displacement the heavy-tailed families lose two orders (1.7e-02). The worst forward-message boundary occupancy is $6.6e-05$ and the worst belief edge occupancy $2.9e-03$ over 60 production cells.

Refinement on the reported estimands. [PENDING: E04 pending: movement of the headline cells under refinement]

Floors in the theory verification. The outer-product information estimate of a shape coordinate is a small residual of order-one quantities; for the kinked Laplace density its quadrature floor is reached near $t = 2.5$ at $N_{\mathrm{g}} = 801$ (the replicate spread stops shrinking), which is why Figure 2b reports the asymptotic slope on the smooth members of the family. The closure gap of the uniform family reaches the double-precision floor of the BP means near $t = 3.5$, where its squared gap is 10^{-13} per chain.

C. Experimental protocol and provenance

Frozen configuration. Laplace chain unless stated, $\rho = 0.85$, $q = 1 - \rho^2$, $a_1 \sim \mathcal{N}(0, 1)$, $L = 32$; grid $N_g = 401$, $A = 8$; twelve log-spaced levels $t \in [0.05, 3.0]$; other families Gaussian, Student-t ($\nu = 5$), uniform, symmetric two-component mixture ($\kappa = 0.6$), generalised Gaussian ($\beta \in [1, 6]$), all variance-matched. Sixteen confirmatory seeds 0–15; development seeds 10000–10003 for every architecture and hyperparameter choice. Validation and test bundles are drawn per seed (1024 and 2048 sequences) from streams disjoint from every training stream; their hashes are stored with every row so pairing across arms is checkable.

Estimand. Schedule-pooled relative score error $\left(\sum_t \sum_{j,i} (\hat{s}_t(\mathbf{x}_j)_i - s_t(\mathbf{x}_j)_i)^2 / \sum_t \sum_{j,i} s_t(\mathbf{x}_j)_i^2\right)^{1/2}$ on all sites (the headline), on the interior sites 5–28 (used for selection), and on the centre site (diagnostic). Ratios are formed within seed and averaged; intervals are percentile bootstraps over seeds.

Arms. EM–BP: mixture-innovation kernel, C as stated, ρ initialised at 0.3, one noised view per sequence with sequences split evenly across the twelve levels, checkpoint ladder $\{10, 20, 40, 60, 80, 120, 160, 220, 300, 400\}$ selected on the validation bundle. Networks: trained by denoising score matching on the clean sequences with the level and the corruption redrawn at every step, batch 64, Adam, gradient clipping at 1, both output parameterisations, checkpoint ladder to 32,000 steps selected on the validation bundle; the information-matched arm trains by implicit score matching on the same one-view noisy bundle as EM–BP and selects on the implicit objective of a held-out noisy bundle.

Provenance. Every run directory carries the commit, the dirty-tree list on the code path, the resolved configuration digest, the seeds, host and environment; a generator refuses to emit a table from a run whose code path was dirty, from an incomplete design, or from cells whose narrowest fitted component is under two grid cells. Every number in this paper is a macro emitted by such a generator; the manuscript check (tools/check_manuscript.py) fails the build on any typed decimal.

D. The oracle ladder

Table 1. Schedule-pooled relative score error of the two oracles against the tree-exact reference, 512 chains per family at the frozen configuration. The LMMSE column is the floor of every second-order estimator (Proposition 1).

family	excess kurtosis	isotropic $-\mathbf{x}$	LMMSE, true $\mathbf{\Sigma}_0$
Laplace	+3	0.670	0.168
Student-t_5	+6	0.668	0.158
uniform	−1.2	0.671	0.149
bimodal	−0.72	0.663	0.082
Gaussian	0	0.662	0.000

E. Capacity: the previous reading and the stopping-rule artefact

[PENDING: The exp_32 protocol (validation on initialisation only; tolerance stop firing at $C = 1$ in 31/32 cells and never at $C \geq 4$), and the E02 table under identical checkpoint selection.]

F. Additional tables

[PENDING: Aggregation robustness; rates; screening details.]